

Latency Minimization for Uplink RSMA System

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Abstract

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A. Uplink Blocklength Minimization for Reduction of Asynchronicity

For MIMO uplink (UL), the receive signal is written as

$$\mathbf{y} = \sum_{k=1}^K \mathbf{H}_k \mathbf{F}_k \mathbf{s}_k + \mathbf{w} \in \mathbb{C}^{N_R},$$

where $\mathbf{H}_k \in \mathbb{C}^{N_R \times N_k}$ denotes the uplink channel between the k th user and the AP, $\mathbf{F}_k \in \mathbb{C}^{N_k \times d_k}$ denotes the precoding matrix for the k th user, d_k denotes the number of spatial streams user k is sending to the AP (and $d_k \leq N_k$), $\mathbf{s}_k \in \mathbb{C}^{d_k}$ denotes the transmitted signal for the k th user, $\mathbf{w} \in \mathbb{C}^{N_R}$ denotes the AWGN, with N_k denoting the number of antennas at the k th user and N_R denotes the number of antennas at the AP. It is assumed that $E[\mathbf{s}_k \mathbf{s}_k^H] = \mathbf{I}_{N_k}$.

For the purpose of our current model, we will assume the case where number of users $K = 2$ and number of spatial stream $d_k = 1$.

Let

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{s}_k,$$

This work has been partially supported by Google exploreCSR Grant 111Q90004C.

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so

$$\begin{aligned} E[\|\mathbf{x}_k\|_2^2] &= \mathbf{F}_k E[\mathbf{s}_k \mathbf{s}_k^H] \mathbf{F}_k^H \\ &= \mathbf{F}_k \mathbf{F}_k^H \\ &= \|\mathbf{F}_k\|_F^2 \leq P_k, \end{aligned}$$

where P_k is the transmit power from the k th user. The instantaneous Shannon rate at the k th receiver can be defined as

$$R_k = \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_k \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{j \neq k} \mathbf{H}_j \mathbf{F}_j \mathbf{F}_j^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_k^H \mathbf{H}_k^H \right)$$

The latency of the k th user can then be written as

$$T_k = \frac{n_k}{f_s}$$

where T_k is the latency, integer n is the channel uses and f_s is the symbol rate.

Considering error probability ϵ_k of the k th user, then the probability and expected value of n_k retransmissions are

$$\begin{aligned} P(n \text{ transmissions}) &= \epsilon_k^{n-1} (1 - \epsilon_k). \\ \mathbb{E}(\text{no. of transmissions}) &= n_{mean} = \frac{1}{1 - \epsilon_k}, \end{aligned}$$

so that the latency of the k th user becomes

$$T_k = \frac{n_k}{f_s} \times \frac{1}{1 - \epsilon_k}.$$

Since a finite blocklength n_k is considered for the k th user, the effective rate of the k th user is

$$R_k^*(n, \epsilon) = R_k - \sqrt{\frac{V_k}{n_k}} Q^{-1}(\epsilon_k),$$

where B_k is the fixed number of bits that need to be sent by k th user and n_k is the number of blocklength of the k th user or number of channels use (or symbols) of the k th user, thus $R_k^*(n, \epsilon)$ is defined in the unit of bits/channel use. Then the latency minimization problem can be formulated as

$$\begin{aligned} \min_{n_k, F_k} \quad & \sum_{k \in \mathcal{K}} w_k \frac{n_k}{f_s} \\ \text{s.t.} \quad & R_k - \sqrt{\frac{V_k}{n_k}} Q^{-1}(\epsilon_k) \geq \gamma_k, \forall k, \\ & n_k \in [n_{\min}, n_{\max}], n_k \in \mathbb{Z}_+, \forall k \\ & \|\mathbf{F}_k\|_F^2 \leq P_k, \forall k, \end{aligned} \tag{1}$$

where w_k is a weight factor for the k th user to determine how much emphasis to place on optimizing its latency, γ_k is the rate threshold of the k th user and $\mathcal{K} \triangleq \{1, 2, \dots, K\}$, with K denoting the maximum number of uplink users.

If retransmissions are also considered, which affects the latency, then (2) can be reformulated as

$$\begin{aligned}
& \min_{n_k, F_k} \sum_{k \in \mathcal{K}} w_k \frac{n_k}{f_s} \frac{1}{1 - \epsilon_k} \\
& s.t. \ R_k - \sqrt{\frac{V_k}{n_k}} Q^{-1}(\epsilon_k) \geq \gamma_k, \forall k, \\
& \quad n_k \in [n_{\min}, n_{\max}], n_k \in \mathbb{Z}_+, \forall k \\
& \quad \epsilon_k \in [0, \epsilon_0], \forall k \\
& \quad \|\mathbf{F}_k\|_F^2 \leq P_k, \forall k.
\end{aligned} \tag{2}$$

Thoughts on the system of equations,

- 1) Reducing n_k reduces the latency $\frac{n_k}{f_s}$.
- 2) Reducing n_k increases the rate $R^*(n, \epsilon)$, however it restricts the upper bound that $R^*(n, \epsilon)$ can achieve by increasing the penalty term $\lim_{n_k \rightarrow 0} \sqrt{\frac{V_k}{n_k}} Q^{-1}(\epsilon_k) = \infty$. Thus n_k must be optimized carefully to keep the upper bound high and come close to achieve it.
- 3) Since latency is the biggest factor of performance and fairness in AUFL, our problem purely on re-balancing the latency to reduce asynchronality. This allows us to change the R_k^* and ϵ_k in a large degree to balance the rate among the users. I believe this is the novelty as i have not come across papers that do this, especially if we optimise the F_k .
- 4) Considering ϵ_k in our optimisation has a huge benefit as there would be exceptionally strong users for whom re-transmission time would be very low. Thus re-transmission would be desired to increase $T_{\text{strong-user}}$ and match the $T_{\text{weak-user}}$ thereby reducing asynchronality. This will allow also us to achieve very high rates for these strong users.

Everything after this is the same as the Review 12-8-25 file.

Possible extension include optimizing not with the sum rate, but using rate such as

1. Max-min rate:

$$\max_{\mathbf{F}_k, \forall k} \min_k R_k.$$

This has a strong fairness guarantee, often leads to equal rates across all users if we can find a feasible solution. This is the worst-case that Carrson was talking about.

2. The proportional fairness:

$$\max_{\mathbf{F}_k, \forall k} \sum_{k=1}^K \log(\mathbf{R}_k),$$

which balances the throughput and fairness.

3. Weighted fairness

$$\max_{\mathbf{F}_k, w_k, \forall k} w_k \log(R_k),$$

where w_k are weights for QoS differentiation, i.e., higher w_k is, more resource will be allocated the k th user.

B. Uplink Latency Minimization Using RSMA

Using RSMA for uplink transmission can theoretically achieve the optimal rate region according to [1] by splitting the original K input channels to $2K - 1$ inputs (though the K th user message need not be split most papers consider splitting the last user message for simplicity).

At user- k the transmitted message of x_k is split into:

$$x_k = \sum_{i=1}^2 F_k \mathbf{s}_{ki},$$

The receive signal at the BS can be written by,

$$\mathbf{y} = \sum_{k=1}^K \sum_{i=1}^2 \mathbf{H}_k \mathbf{F}_{ki} \mathbf{s}_{ki} + \mathbf{w} \in \mathbb{C}^{N_R},$$

where

$$\sum_{i=1}^2 \|\mathbf{F}_{ki}\|_F^2 \leq P_k$$

While decoding at the BS SIC is used, this involves the use of a decoding order π , let the permutation π belong to the set Π defined as the set of all possible decoding order of all $2K$ messages from K users.

Let the decoding order of message s_{ki} from the user- k be π_{ki} , then achievable rate of decoding message s_{ki} can be given as,

Let $\mathbf{K}$ be the set of user devices, Let $\mathbf{J}$ be the number of split streams $\{1, 2\}$

$$R_{ki} = \log_2 \det \left(\mathbf{I}_{N_R} + \mathbf{H}_k \mathbf{F}_{ki} \left(\sigma_w^2 \mathbf{I}_{N_R} + \sum_{\{(j \in \mathbf{K}, i \in \mathcal{J}) | \pi_{ji} > \pi_{ki}\}} \mathbf{H}_j \mathbf{F}_{ji} \mathbf{F}_{ji}^H \mathbf{H}_j^H \right)^{-1} \mathbf{F}_{ki}^H \mathbf{H}_k^H \right)$$

The set $\{(j \in \mathcal{K}, i \in \mathcal{J}) | \pi_{ji} > \pi_{ki}\}$ represent the messages s_{ji} that are decoded after message s_{ki} .

$$R_k = \sum_{i=1}^2 R_{ki}$$

To optimize for the rate in Uplink RSMA, a joint optimization of both the power management and message decoding order must be done.

Max-Min Rate

$$\begin{aligned} & \max_{\mathbf{F}_k, \forall k} \min_{k, \pi} R_k. \\ \text{s.t. } & \sum_{i=1}^2 \|\mathbf{F}_{ki}\|_F^2 \leq P_k \\ & \pi \in \Pi, \|\mathbf{F}_{ki}\|_F^2 \geq 0, \forall k \in \mathbf{K}, j \in \mathbf{J} \end{aligned}$$

This is similar to the optimization problem modeled in [2] where they optimize for the sum rate.

C. Uplink Latency Minimization Using Finite Blocklength Minimization

To consider latency in URRLC systems with finite blocklength, let us model channel rate not in terms of (bits/s) but as (bits/channel use).

Let blocklength = n be the (channel uses), and let T_s be the time taken to transmit one symbol.

Then,

$$\begin{aligned} \text{Latency} &= n \times T_s \\ &= \frac{B(\text{bits})}{R(\text{bits/channel use})} \times T_s(\text{sec}) \end{aligned}$$

And thus latency is optimised for by optimising for n (blocklength).

Rate under Finite Blocklength according to [3],

$$R^*(n, \epsilon) = C - \sqrt{\frac{V}{n}} Q^{-1}(\epsilon) + \mathcal{O}\left(\frac{\log n}{n}\right)$$

where C is Shannon capacity of channel, $Q^{-1}(\cdot)$ is the inverse of the Gaussian Q function, V is the channel dispersion.

This rate equation according to [3] says that, to sustain the desired error probability ϵ for a given packet size n , one incurs a penalty on the rate (compared to the channel capacity) that is proportional to $\frac{1}{\sqrt{n}}$

Rate under Finite Blocklength in Massive MIMO systems according to [4],

Rate in high SNR regime is approximated as,

$$R_k \approx d_k \log(1 + \rho_k) - \sqrt{\frac{d_k}{n_k}} Q^{-1}(\epsilon_k).$$

(I am not sure why they removed the variance V .)

R_k is rate of user-k, ρ_k is the SNR of user-k and $Q^{-1}(\cdot)$ denotes the inverse of Gaussian Q function.

$$Q^{-1}(x) \triangleq \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

Re-arranging the equation, when d_k is large enough,

$$n_k \approx \frac{d_k [Q^{-1}(\epsilon_k)]^2}{[d_k \log_2(1 + \rho_k) - R_k]^2} \propto \frac{1}{d_k},$$

This shows that the latency in the time domain can be reduced by increasing the DoFs in the space domain, for the purpose of maintaining a certain level of performance target. [4]

$$n_k = B_k / R_k$$

$$n_k = \frac{B_k}{d_k \log(1 + \rho_k) - \sqrt{\frac{d_k}{n_k}} Q^{-1}(\epsilon_k)}.$$

Solving this equation would give another equation that describe n_k if we know the bits B_k . I am not sure how this fits in with the other formulas.

From the equations it can also be seen that to decrease n_k , we can:

- 1) Increase spacial DoF d_k
- 2) Increase SNR ρ_k
- 3) Increase error probability ϵ_k (reliability)

Min-Max Blocklength/Latency

$$\min_{\forall k} \max_{k, R_k, \epsilon_k} n_k.$$

D. Uplink Latency Minimization with RSMA and Finite Blocklength Minimization

The paper to read on this topic is [5], it deals with almost the exact same problem setup as us.

They consider minimizing the blocklength in Uplink RSMA in a simple 2 user system.

However in our setup, since we want to improve the latency of the worst performing user, we should consider increasing: d_k , ρ_k , ϵ_k of the weakest user through precoder design etc, while decreasing the same parameters of the stronger users to increase their blocklength thus their latency and reduce asynchronality.

I feel this has a naturally synergistic approach, as even though you increase the error probability of the weaker user, the weighing mechanism in Fed-async takes care to reduce the impact of data from these users on the training.

Similarly reducing the error probability of the stronger user will be helpful as the data from the stronger user is weighed higher.

This helps reduce the unfairness of asynchronality and ensure that the weaker users still have an impact on the training while taking care not to weigh it too highly due to their high error probability.

This is something that I do not believe is explored in [5].

REFERENCES

- [1] B. Rimoldi and R. Urbanke, "A rate-splitting approach to the Gaussian multiple-access channel," *IEEE Trans. Inf. Theory*, vol. 42, no. 2, pp. 364–375, Mar. 1996.
- [2] Z. Yang, M. Chen, W. Saad, W. Xu and M. Shikh-Bahaei, "Sum-Rate Maximization of Uplink Rate Splitting Multiple Access (RSMA) Communication," 2019 IEEE Global Communications Conference (GLOBECOM), Waikoloa, HI, USA, 2019, pp. 1-6, doi: 10.1109/GLOBECOM38437.2019.9013344.
- [3] G. Durisi, T. Koch and P. Popovski, "Toward Massive, Ultrareliable, and Low-Latency Wireless Communication With Short Packets," in *Proceedings of the IEEE*
- [4] X. You et al., "Closed-Form Approximation for Performance Bound of Finite Blocklength Massive MIMO Transmission," in *IEEE Transactions on Communications*, vol. 71, no. 12, pp. 6939-6951, Dec. 2023

- [5] Y. Zhang, W. Cheng, J. Wang and W. Zhang, "Performance Analysis and Blocklength Minimization of Uplink RSMA for Short Packet Transmissions in URLLC," GLOBECOM 2023 - 2023 IEEE Global Communications Conference, Kuala Lumpur, Malaysia, 2023, pp. 6765-6770